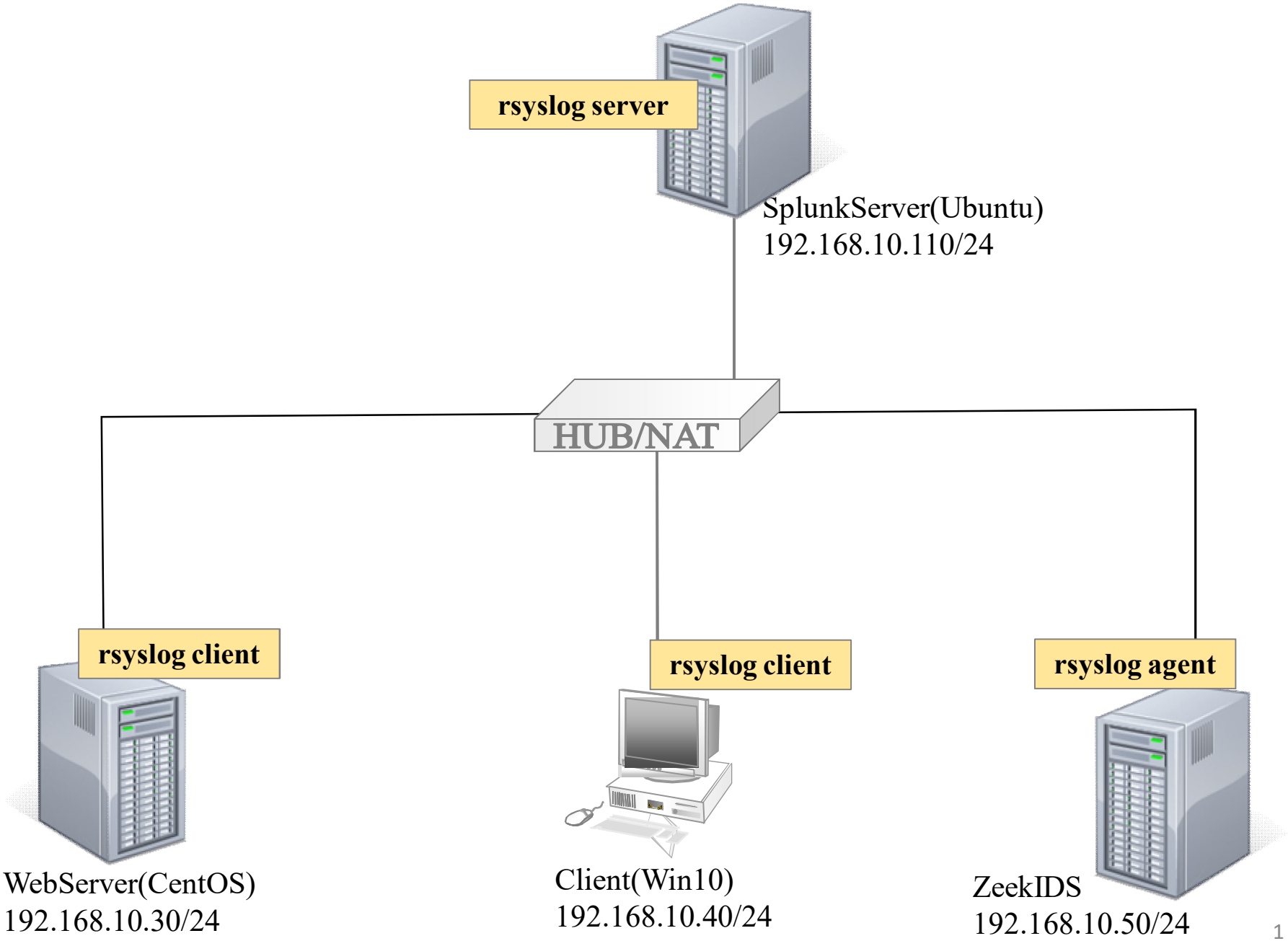

rsyslog server
SplunkServer(Ubuntu)
192.168.10.110/24
HUB/NAT
rsyslog client
WebServer(CentOS)
192.168.10.30/24
rsyslog client
Client(Win10)
192.168.10.40/24
rsyslog agent
ZeekIDS
192.168.10.50/24

1) [Splunk server] rsyslog server installation

❶ #apt-get install –y rsyslog

❷ #nano /etc/rsyslog.config

❸ #ufw allow 514/tcp

❹ #ufw reload

❺ #systemctl restart rsyslog

❶ #apt-get install –y rsyslog

❷ #nano /etc/rsyslog.config

```
root@splunk-server:/# apt-get install -y rsyslog
패키지 목록을 읽는 중입니다... 완료
의존성 트리를 만드는 중입니다
상태 정보를 읽는 중입니다... 완료
패키지 rsyslog는 이미 최신 버전입니다 (8.2001.0-1ubuntu1).
rsyslog 패키지는 수동설치로 지정합니다.
0개 업그레이드, 0개 새로 설치, 0개 제거 및 0개 업그레이드 안 함.
root@splunk-server:/#
```

❷ #nano /etc/rsyslog.config

```
root@splunk-server:/# cat /etc/rsyslog.conf
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#
# Default logging rules can be found in /etc/rsyslog.d/50-default.conf


#################
#### MODULES ####
#################

module(load="imuxsock") # provides support for local system logging
#module(load="immark")  # provides --MARK-- message capability

# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")

# provides TCP syslog reception
module(load="imtcp")
input(type="imtcp" port="514")
```

❷ #nano /etc/rsyslog.config

```
root@splunk-server:/# tail /etc/rsyslog.conf
#
$WorkDirectory /var/spool/rsyslog

#
# Include all config files in /etc/rsyslog.d/
#
$IncludeConfig /etc/rsyslog.d/*.conf

$template remote-incoming-logs, "/var/log/192.168.10.0/client.log
*.*?remote-incoming-logs
root@splunk-server:/#
```

$template remote-incoming-logs,"/var/log/%HOSTNAME%/%PROGRAMNAME%.log"
**.* ?remote-incoming-logs*

클라이언트호스트 생성된 로그 파일명

```
$template remote-incoming-logs, "/var/log/192.168.10.0/client.log
*.*?remote-incoming-logs
```

❸ 방화벽 활성화

#ufw allow 514/tcp

#ufw reload

```
root@splunk-server:/# ufw allow 514/tcp
규칙이 추가되었습니다
규칙이 추가되었습니다 (v6)
root@splunk-server:/# ufw allow 514/udp
규칙이 추가되었습니다
규칙이 추가되었습니다 (v6)
root@splunk-server:/# ufw status
상태: 활성

목적                       동작         출발
--                       --         --
123/udp                  ALLOW      Anywhere
514/tcp                  ALLOW      Anywhere
514/udp                  ALLOW      Anywhere
123/udp (v6)             ALLOW      Anywhere (v6)
514/tcp (v6)             ALLOW      Anywhere (v6)
514/udp (v6)             ALLOW      Anywhere (v6)

root@splunk-server:/#
```

❹ 서비스 활성화

#systemctl restart rsyslog

#systemctl status rsyslog

```
root@splunk-server:/# systemctl restart rsyslog
root@splunk-server:/# systemctl status syslog
●rsyslog.service - System Logging Service
     Loaded: loaded (/lib/systemd/system/rsyslog.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2022-08-07 00:24:25 KST; 5s ago
TriggeredBy: ●syslog.socket
       Docs: man:rsyslogd(8)
             https://www.rsyslog.com/doc/
   Main PID: 4941 (rsyslogd)
      Tasks: 10 (limit: 9420)
     Memory: 1.4M
     CGroup: /system.slice/rsyslog.service
             └─4941 /usr/sbin/rsyslogd -n -iNONE

 8월 07 00:24:25 splunk-server systemd[1]: Starting System Logging Service...
 8월 07 00:24:25 splunk-server rsyslogd[4941]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from >
 8월 07 00:24:25 splunk-server rsyslogd[4941]: rsyslogd's groupid changed to 110
 8월 07 00:24:25 splunk-server systemd[1]: Started System Logging Service.
 8월 07 00:24:25 splunk-server rsyslogd[4941]: rsyslogd's userid changed to 104
 8월 07 00:24:25 splunk-server rsyslogd[4941]: [origin software="rsyslogd" swVersion="8.2001.0" x-pid="4941" x-info="htt>
lines 1-18/18 (END)
```

❺ 생성된 로그 디렉터리와 로그 확인

#cd /var/log/192.168.110.0

```
root@splunk-server:/var/log# ls -l
합계 4248
drwxr-xr-x  2 syslog            syslog            4096  8월  7 01:10 192.168.10.0
-rw-r--r--  1 root              root             21158  7월 30 11:24 alternatives.log
```

```
root@splunk-server:/var/log/192.168.10.0# pwd
/var/log/192.168.10.0
root@splunk-server:/var/log/192.168.10.0# ls
client.log
root@splunk-server:/var/log/192.168.10.0# tail client.log
Aug  7 01:11:36 localhost journal[3031]: failed to get featured apps: no apps to show
Aug  7 01:11:36 localhost journal[3031]: not handling error failed for action refine: failed to resolv
Aug  7 01:11:36 localhost PackageKit[2926]: resolve transaction /101_cdadcbeb from uid 0 finished wit
Aug  7 01:11:36 localhost journal[3031]: Only 0 apps for popular list, hiding
Aug  7 01:11:36 localhost journal[3031]: not handling error failed for action refine: failed to resolv
Aug  7 01:11:37 localhost PackageKit[2926]: get-updates transaction /102_cbdbcccb from uid 0 finished
Aug  7 01:12:31 splunk-server ntpd[4033]: Soliciting pool server 2620:2d:4000:1::3f
Aug  7 01:13:38 splunk-server ntpd[4033]: Soliciting pool server 2620:2d:4000:1::41
Aug  7 01:14:44 splunk-server ntpd[4033]: Soliciting pool server 2620:2d:4000:1::41
Aug  7 01:15:50 splunk-server ntpd[4033]: Soliciting pool server 2620:2d:4000:1::3f
root@splunk-server:/var/log/192.168.10.0#
```

2) [Ubuntu-Zeek] rsyslog client installation

❶ #apt-get install –y rsyslog

❷ #nano /etc/rsyslog.conf

```
*.notice        @@192.168.10.10:514
```

시스템에서 발생하는 모든 종류의 로그를 192.168.10.110:514로 TCP를 이용해서 전송

❸ systemctl status rsyslog

❹ #ufw allow 514/tcp

❺ #ufw reload

❻ #systemctl restart rsyslog

❶ #apt-get install –y rsyslog

```
root@zeek-IDS:~#
root@zeek-IDS:~# dpkg -l rsyslog
희망상태=알수없음(U)/설치(I)/지우기(R)/깨끗이(P)/고정(H)
| 상태=아님(N)/설치(I)/설정(C)/풀림(U)/절반설정(F)/일부설치(H)/트리거대기(W)/
| /    트리거밀림(T)
|/ 오류?=(없음)/다시설치필요(R) (상태, 오류가 대문자=불량)
||/ 이름           버전              Architecture 설명
+++-==============-=================-============-=======================================
ii  rsyslog        8.2001.0-1ubuntu1 amd64        reliable system and kernel logging daemon
root@zeek-IDS:~#
```

❷ #nano /etc/rsyslog.conf

```
root@zeek-IDS:~# tail /etc/rsyslog.conf

#
# Include all config files in /etc/rsyslog.d/
#
$IncludeConfig /etc/rsyslog.d/*.conf

*.notice        @@192.168.10.10:514



root@zeek-IDS:~#
```

3) [CentOS] rsyslog client installation

❶ #rpm –qa | grep rsyslog

❷ #gedit etc/rsyslog.conf

❸ # systemctl restart rsyslog

❹ #firewall-config

❶ #apt-get install –y rsyslog

```
[root@localhost ~]# rpm -qa | grep rsyslog
rsyslog-gnutls-8.37.0-9.el8.x86_64
rsyslog-8.37.0-9.el8.x86_64
rsyslog-relp-8.37.0-9.el8.x86_64
rsyslog-gssapi-8.37.0-9.el8.x86_64
```

❷ #nano /etc/rsyslog.conf

```
[root@localhost ~]# tail -15 /etc/rsyslog.conf

*.notice                                                @@192.168.10.10:514

# ### sample forwarding rule ###
#action(type="omfwd"
# An on-disk queue is created for this action. If the remote host is
```

*.notice @@192.168.10.110:514

❸ 서비스 활성화

#systemctl restart rsyslog

#systemctl status rssylog

systemctl enable rsyslog

```
[root@localhost ~]# systemctl restart rsyslog
[root@localhost ~]# systemctl status rsyslog
● rsyslog.service - System Logging Service
   Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2022-08-07 00:44:54 KST; 10s ago
     Docs: man:rsyslogd(8)
           http://www.rsyslog.com/doc/
 Main PID: 3972 (rsyslogd)
    Tasks: 3 (limit: 24877)
   Memory: 1.2M
   CGroup: /system.slice/rsyslog.service
           └─3972 /usr/sbin/rsyslogd -n

[root@localhost ~]# systemctl enable rsyslog
```

#netstat –natlp | grep rsyslog

#lsof –i tcp 514

```
[root@localhost ~]# netstat -natlp | grep rsyslog
tcp        0      0 192.168.10.30:40834     192.168.10.10:514       ESTABLISHED 3972/rsyslogd
[root@localhost ~]#
[root@localhost ~]# lsof -i tcp:514
COMMAND   PID USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
rsyslogd 3972 root    7u  IPv4  67014      0t0  TCP localhost.localdomain:40834->splunk-server:shell (ESTABLISHED)
```

❹ 방화벽 설정

iptables -A INPUT -m state --state NEW -m tcp -p tcp --dport 514 -j ACCEPT

iptables -A INPUT -m state --state NEW -m udp -p udp --dport 514 -j ACCEPT

firewall-cmd --permanent --add-port=514/tcp

firewall-cmd --permanent --add-port=514/udp

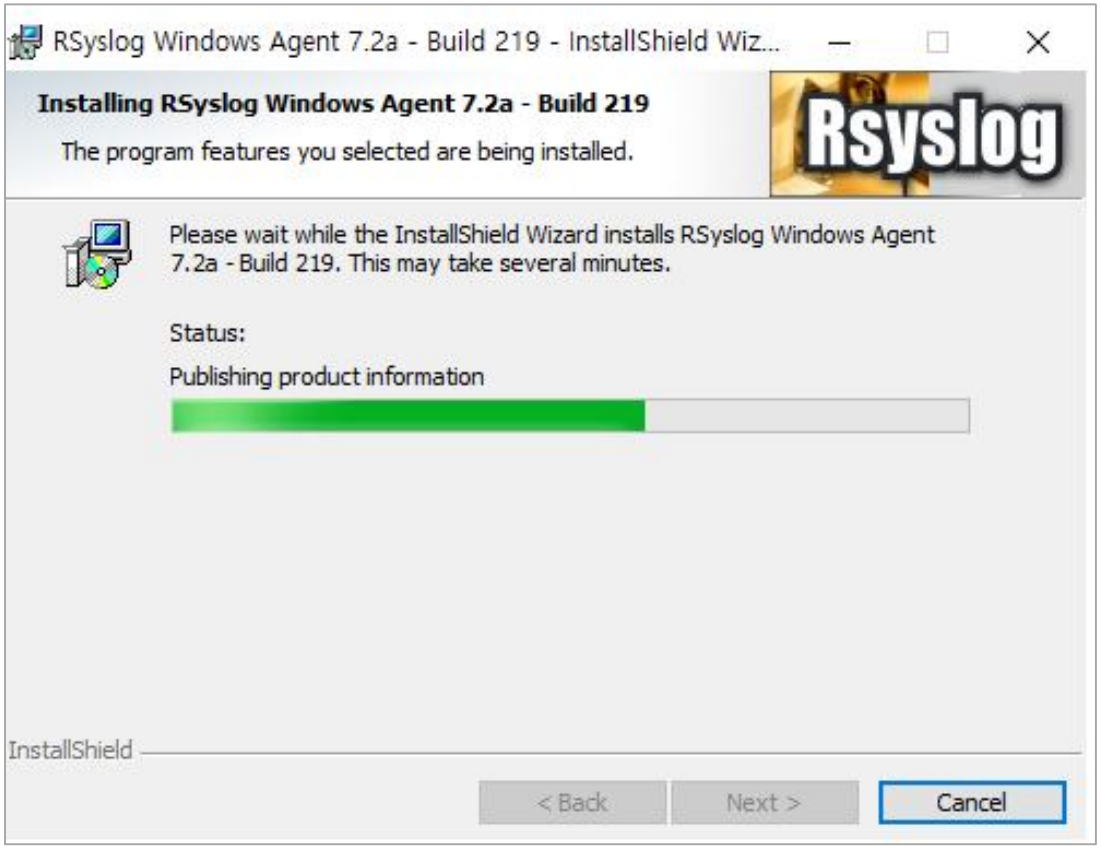
firewall-cmd –reload

firewall-cmd --list-ports

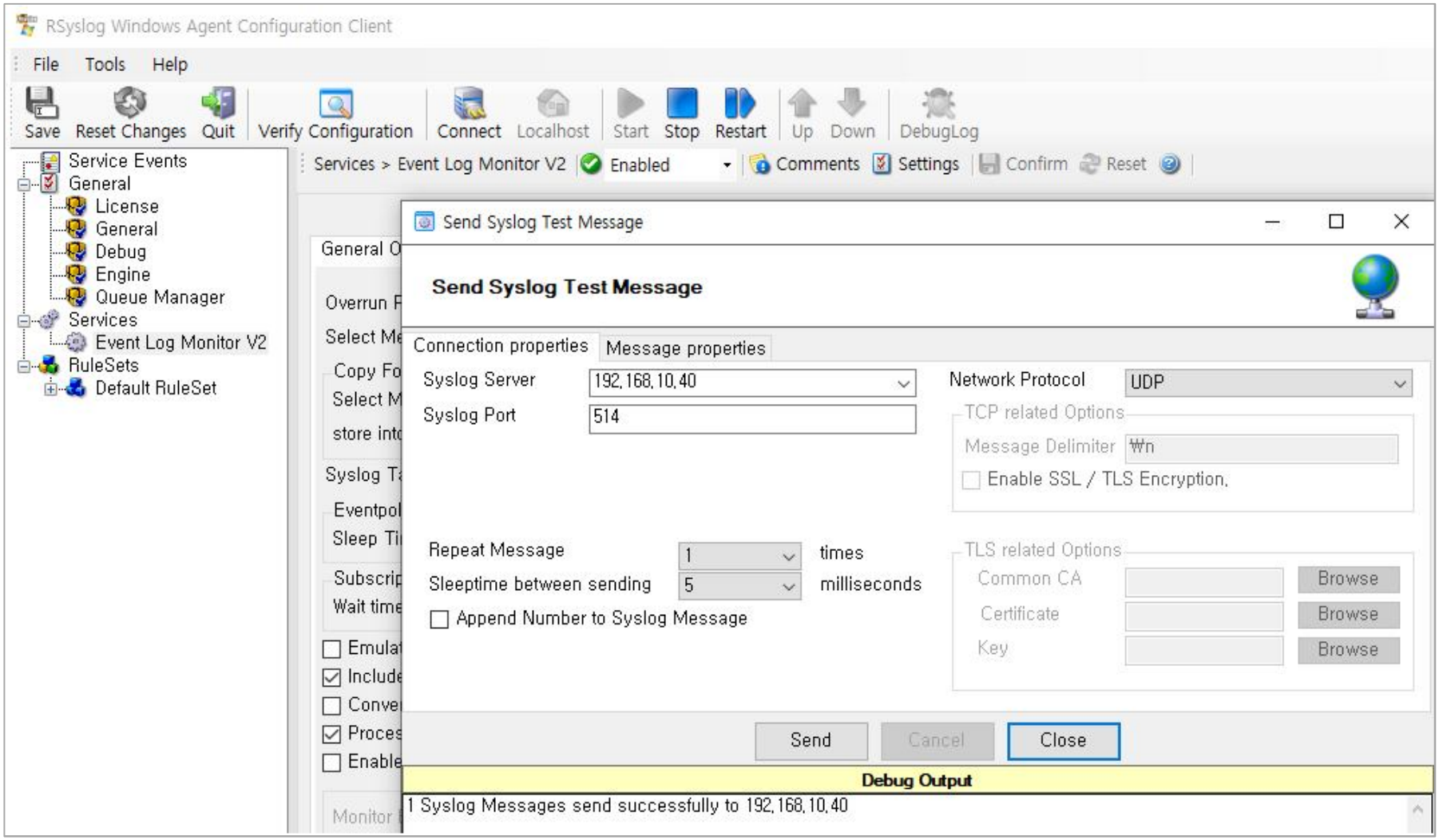
```
[root@localhost ~]# iptables -A INPUT -m state --state NEW -m tcp -p tcp --dport 514 -j ACCEPT
[root@localhost ~]# iptables -A INPUT -m state --state NEW -m udp -p udp --dport 514 -j ACCEPT
[root@localhost ~]# firewall-cmd --permanent --add-port=514/tcp
success
[root@localhost ~]# firewall-cmd --permanent --add-port=514/udp
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-ports
514/tcp 514/udp
[root@localhost ~]#
```

4) [Win10] rsyslog client installation

https://www.rsyslog.com/windows-agent/



Rsyslog Windows Agent


RSyslog Windows Agent Configuration Client
File
Tools
Help
Save
Reset Changes
Quit
Verify Configuration
Connect
Localhost
Start
Stop
Restart
Up
Down
DebugLog
Service Events
General
License
General
Debug
Engine
Queue Manager
Services
Event Log Monitor V2
RuleSets
Default RuleSet
Services > Event Log Monitor V2
Enabled
Comments
Settings
Confirm
Reset
Send Syslog Test Message
Send Syslog Test Message
Connection properties
Message properties
Syslog Server
192.168.10.40
Syslog Port
514
Network Protocol
UDP
TCP related Options
Message Delimiter
₩n
Enable SSL / TLS Encryption.
Repeat Message
1
times
Sleeptime between sending
5
milliseconds
Append Number to Syslog Message
TLS related Options
Common CA
Browse
Certificate
Browse
Key
Browse
Send
Cancel
Close
Debug Output
1 Syslog Messages send successfully to 192.168.10.40